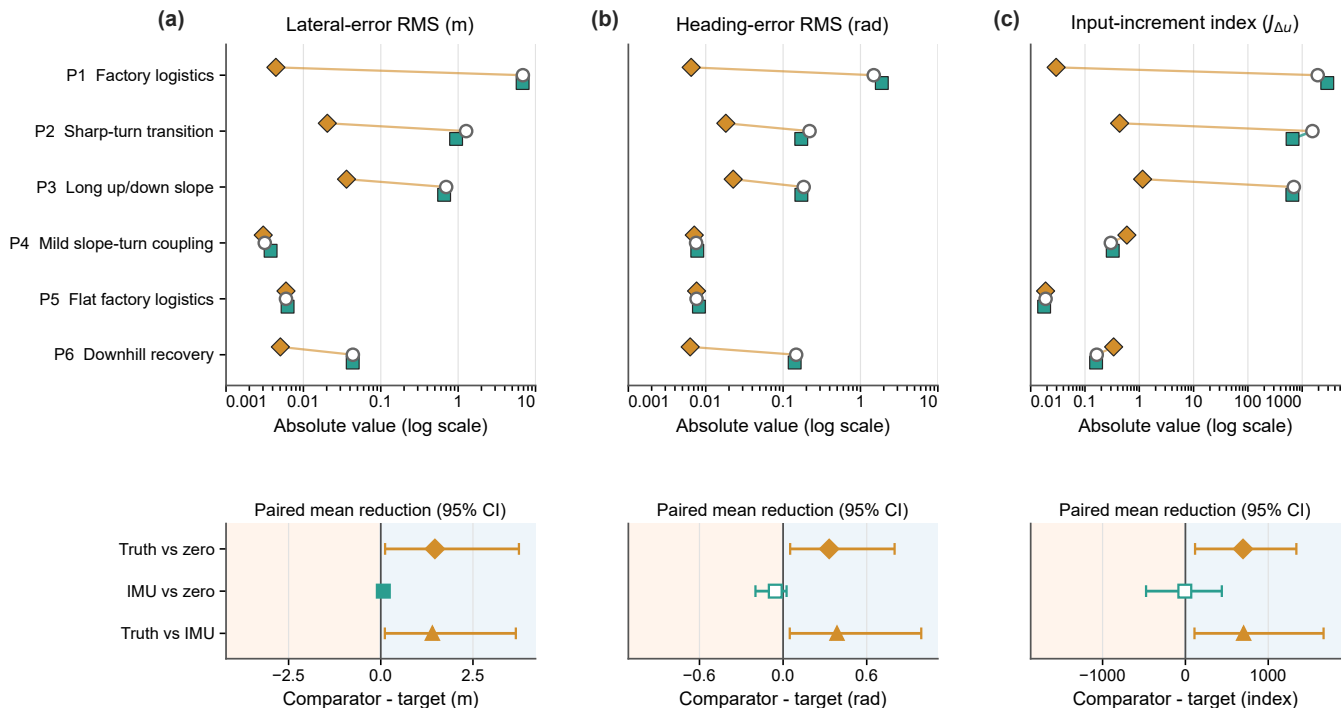


○ Zero-grade (2.781 deg MAE)    ■ Causal-IMU (2.728 deg MAE)    ◆ Truth-driven (0.123 deg MAE; analysis-only)

Grade-source accuracy across six routes; lower scheduled-grade MAE is more accurate.



$n = 6$  paired routes; whiskers are 95% route-bootstrap CIs. Positive mean reduction = lower/better target.  
Filled summary markers have CIs excluding zero; route-level adverse and tied outcomes are retained.